
Robinhood: Welcome to the New Wall Street?

Through Robinhood, millions of everyday people have started investing in the stock market for the first time. They're nurses, Army veterans, teachers, students, and small business owners...The new Wall Street lives on Main Street, and it's a place for everyone.

—Robinhood Newsroom¹

Serena Khan, an analyst at a multibillion-dollar asset management firm, was heading back to her office from lunch with an investor when her phone buzzed. She clicked a button, and the voice of her portfolio manager boss, Dennis Moore, suddenly boomed through her BMW speakers. He quickly got to the point.

“Serena, I want you to take a look at Robinhood. I’ve got this stock on my screen again. It’s even cheaper than the last time I looked at it.”

Khan, who had earned her MBA the previous year, had long admired Robinhood Market, Inc.’s (Robinhood), mission to democratize finance by offering commission-free trading via its user-friendly app. It appeared, however, that other investors had largely soured on this model. By March 2023, Robinhood’s stock had dropped below \$10, a decline of over 80% from its peak less than two years earlier. “My gut worries that it could be a value trap,” she responded.

“It could be. But it could be a great opportunity. I’m trying to wrap my head around its business model. The company has taken a lot of heat for how it makes money and its role in the GameStop fiasco. What are the risks associated with the model? Was this company just a flash in the pan, or is there a path toward sustainable growth? Can it take on its competitors in the brokerage industry?”

“Okay. I can have a look. When do you want this done?” asked Khan.

“Can you come back to me in two days?”

¹ “Welcome to the New Wall Street,” Robinhood Newsroom, July 29, 2021, <https://newsroom.aboutrobinhood.com/welcome-to-the-new-wall-street/> (accessed Feb. 5, 2024).

This public-sourced case was prepared by Ian Appel, Associate Professor of Business Administration, Ting Xu, Assistant Professor of Finance (University of Toronto), and Stephen E. Maiden (MBA '01), Senior Case Researcher. It was written as a basis for class discussion rather than to illustrate effective or ineffective handling of an administrative situation. Serena Khan and Dennis Moore were fictionalized for pedagogical reasons. Copyright © 2024 by the University of Virginia Darden School Foundation, Charlottesville, VA. All rights reserved. *To order copies, send an email to sales@dardenbusinesspublishing.com. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of the Darden School Foundation. This publication is protected by copyright and may not be uploaded in whole or part to any AI, large language model, or similar system, or to any related training database.* Our goal is to publish materials of the highest quality, so please submit any errata to editorial@dardenbusinesspublishing.com.

The Brokerage Industry

Khan got to work as soon as she returned to the office. As was typical of her approach, she began with an overview of the industry. Khan recognized the critical role brokers played in financial markets by executing transactions on behalf of clients. She was less familiar with how the economics of the brokerage industry had evolved over time.

Commissions: A historical overview

Throughout much of Wall Street's history, brokerage firms operated as a "quasi-cartel" that fixed the price of trading stocks and bonds.² For example, **Table 1** reports the commissions for stock trades on the NYSE between 1959 and 1968. While price-fixing muted competition between brokers, it was viewed as a "sacred creed" of exchanges.³

Table 1. NYSE commission schedule (March 3, 1959, to December 5, 1968).

Money Involved	Commission
\$100 to \$400	\$3 + 2.0% of amount traded
\$400 to \$2,400	\$7 + 1.0% of amount traded
\$2,400 to \$5,000	\$19 + 0.5% of amount traded
Over \$5,000	\$39 + 0.1% of amount traded

Data source: Charles M. Jones, "A Century of Stock Market Liquidity and Trading Costs," Columbia Business School, September 13, 2002, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=313681 (accessed Feb. 6, 2024).

In the 1960s, regulators targeted the NYSE's fixed commissions, citing their anticompetitive nature. Despite industry resistance, fixed commissions were eventually banned effective May 1, 1975—a date known as "Mayday" among brokers.⁴ Institutions and other large investors immediately benefited, with commissions falling by as much as 50%. Retail investors (i.e., individuals with relatively small portfolios) realized fewer benefits, at least initially, because "the big brokerage houses had no interest in dickering over fees for a tiny trade."⁵

New firms, termed discount brokers, emerged to cater to retail investors. A 1990 *New York Times* article noted that this industry was dominated by three players: Charles Schwab & Co. (Schwab), Fidelity Investments, and Quick & Reilly, Inc.⁶ The era of the personal computer brought further changes to the industry and the emergence of new, tech-friendly brokers (e.g., E-Trade Financial Corporation [E-Trade]). Commissions steadily declined throughout the 1980s and 1990s.⁷

In more recent years, commissions continued to be an important aspect of the discount broker business model, making up between 13% and 48% of revenues in 2013 (**Exhibit 2**). The 2014 launch of Robinhood, a

² Stephen Mihm, "The Death of Brokerage Fees Was 50 Years in the Making," Bloomberg, January 3, 2020, <https://www.bloomberg.com/view/articles/2020-01-03/how-nyse-went-from-quasi-cartel-to-zero-fee-stock-trading> (accessed Feb. 5, 2024).

³ <https://www.bloomberg.com/view/articles/2020-01-03/how-nyse-went-from-quasi-cartel-to-zero-fee-stock-trading>.

⁴ <https://www.bloomberg.com/view/articles/2020-01-03/how-nyse-went-from-quasi-cartel-to-zero-fee-stock-trading>.

⁵ <https://www.bloomberg.com/view/articles/2020-01-03/how-nyse-went-from-quasi-cartel-to-zero-fee-stock-trading>.

⁶ Richard D. Hylton, "All About: Discount Brokers; Now Fewer Firms Are Chasing Small Investors Discount Brokers," *New York Times*, June 17, 1990, <https://www.nytimes.com/1990/06/17/business/all-about-discount-brokers-now-fewer-firms-are-chasing-small-investors-discount.html> (accessed Feb. 5, 2024).

⁷ For historical evidence, see Charles M. Jones, "A Century of Stock Market Liquidity and Trading Costs," Columbia Business School, September 13, 2002, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=313681 (accessed Feb. 5, 2024).

zero-commission broker, significantly altered the industry dynamics. Rather than charging commissions, Robinhood's revenue came from the market makers that executed clients' trades, a model known as "payment for order flow" (PFOF). As Robinhood gained traction, its competitors responded. For example, in 2017, Schwab reduced commissions on stock and exchange-traded fund transactions from \$6.95 to \$4.95.⁸ Less than two years later, Schwab cut commissions to zero. Around the same time, Interactive Brokers LLC (Interactive Brokers) and TD Ameritrade announced they were eliminating some commissions as well.⁹

Market makers and PFOF

Historically, most stock trades were executed via public markets. Prior to the late 1960s, for example, to buy 100 shares of a stock, an investor would call a broker who would then transmit the order to traders on the floor of the NYSE. From there, the order would be matched with an order to sell the same number of shares in the company.¹⁰ The 1980s saw the rise of electronic trading, which largely replaced floor traders. Over time, trading moved away from major exchanges. **Exhibit 3** reports venues with the most trading volume in December 2020. Exchanges, including the NYSE and Nasdaq, accounted for less than 60% of volume. The largest non-exchange venues—Citadel Securities LLC (Citadel Securities),¹¹ Virtu Financial Inc., G1 Execution Services LLC, and Two Sigma Investments, LP—were firms that internally matched orders from buyers and sellers.

Market makers made money from the difference between the price they were willing to pay for an asset (the bid) versus the price they were willing to sell at (the ask). In some cases, brokers were able to obtain better pricing by routing orders to market makers rather than exchanges. For example, consider a stock with a bid of \$10.00 and an ask of \$10.05 on an exchange. These prices, which indicated that a trader could sell shares at \$10.00 and buy at \$10.05, were known as the national best bid and offer (NBBO).¹² A market maker, though, might be willing to buy shares for \$10.01 and sell them for \$10.04, thus offering improved pricing relative to the NBBO. Market makers profited from the difference between these prices (i.e., the bid-ask spread). While profits for individual transactions were small, massive volume made up for it. For example, as of the summer of 2022, Citadel Securities had revenues of at least \$1 billion for 10 consecutive quarters.¹³

Market makers vied for brokerage business, often using PFOF as an incentive. Originating in the 1980s, PFOF had recently become a crucial component of many brokers' business models. In 2020 alone, four brokers—Robinhood, TD Ameritrade, E-Trade, and Schwab—had \$2.5 billion in PFOF revenue.¹⁴ Jim Angel, a finance professor at Georgetown University, noted that order flow from these brokers was particularly valuable because they catered to retail investors:

Market makers particularly like to take the other side of small retail trades because they know that those retail traders are not sophisticated institutional investors such as hedge funds. Market makers know that they can lose when they trade with large institutions that know more than they do. Market makers

⁸ "Schwab Reduces Trade Commissions to \$4.95 and Lowers Per Contract Options Fee to \$0.65," Charles Schwab Corporation press release, February 28, 2017, <https://pressroom.aboutschwab.com/press-releases/press-release/2017/Schwab-Reduces-Trade-Commissions-to-4.95-and-Lowers-Per-Contract-Options-Fee-to-0.65/default.aspx> (accessed Feb. 5, 2024).

⁹ Alexander Osipovich, "Schwab Cuts Fees on Online Stock Trades to Zero, Rattling Rivals," *Wall Street Journal*, October 1, 2019, <https://www.wsj.com/articles/charles-schwab-ending-online-trading-commissions-on-u-s-listed-products-11569935983> (accessed Feb. 5, 2024).

¹⁰ Bob Pisani, "Man vs. Machine: How Stock Trading Got So Complex," CNBC, September 13, 2010, <https://www.cnbc.com/id/38978686> (accessed Feb. 5, 2024).

¹¹ Citadel Securities LLC was a separate entity from the hedge fund Citadel LLC, though both were founded by hedge fund manager Kenneth C. Griffin.

¹² Matt Levine, "What Does Payment for Order Flow Buy?" Bloomberg, December 8, 2021, <https://www.bloomberg.com/opinion/articles/2021-12-08/what-does-payment-for-order-flow-buy> (accessed Feb. 5, 2024).

¹³ "Citadel Securities Revenue Surges to \$4.2 Billion on Volatility," Bloomberg, August 16, 2022, <https://www.bloomberg.com/news/articles/2022-08-15/citadel-securities-first-half-trading-revenue-hits-4-2-billion> (accessed Feb. 5, 2024).

¹⁴ "Broker-Dealers and Payment for Order Flow," Congressional Research Service, April 2, 2021, <https://crsreports.congress.gov/product/pdf/IF/IF11800> (accessed Feb. 5, 2024).

can buy from a retail order at the bid and sell at the ask or offer and pocket the bid-ask spread. They don't need to worry that retail investors as a group have better information and will dump shares on the market maker just before bad news is announced.¹⁵

PFOF was a contentious practice. Investor advocates and regulators alike recognized that PFOF incentivized brokers to sell orders to the highest bidder, not necessarily the market maker that offered the best execution price. Indeed, PFOF was prohibited in some countries (e.g., the United Kingdom and Canada) for this very reason.¹⁶ In the United States, PFOF remained legal but was subject to several rules and regulations. Most importantly, brokers were bound by the “duty of best execution,” requiring them to obtain the most favorable prices for customers.¹⁷

Robinhood

Next, Khan turned attention to Robinhood itself. She sought to better understand the company's business model and the factors that had contributed to its early growth.

Business model

Robinhood was founded by Vladimir Tenev and Baiju Bhatt, Stanford University classmates who had previously built trading software used by hedge funds. The duo's goal: democratize finance. Tenev explained:

We fundamentally believe that participation in the U.S. capital markets is empowering and that everyone should have the opportunity to participate responsibly in our financial system. Today, however, approximately 84 percent of the value of all stocks owned by Americans belong to the wealthiest 10 percent of households, and roughly half of all American households do not own any stocks at all. We want to open up access to the markets so everyday people, even those with small amounts to invest, can build wealth.¹⁸

The founders sought to address two main obstacles faced by retail investors. First, most brokers charged flat fees for transactions that made investing relatively small amounts of money impractical. Second, for many, the investing process was intimidating, with many brokers using interfaces designed for seasoned professionals rather than newcomers. In addressing these challenges, Robinhood attracted a customer base that differed from that of other brokerages. The average Robinhood user was 31 years old and had an account size of \$4,500; the average Interactive Brokers customer was more than a decade older and had an account size of \$257,000.¹⁹

Zero commissions

Rather than relying on commissions, Robinhood earned revenue via alternative channels, primarily PFOF. In a 2018 letter, Tenev noted that Robinhood PFOF payments averaged \$0.026 on a \$100.00 transaction. The company routed customers' orders to several market makers, the largest being Citadel Securities (**Table 2**). In

¹⁵ <https://crsreports.congress.gov/product/pdf/IF/IF11800>.

¹⁶ “Brokers Braced for Big Overhaul of US Stock Trading Rules,” *Financial Times*, December 13, 2022, <https://www.ft.com/content/66bb47cb-ab02-4764-90da-6c6c05c53672> (accessed Feb. 5, 2024).

¹⁷ Jonathan R. Macey and Maureen O'Hara, “The Law and Economics of Best Execution,” *Journal of Financial Intermediation* 6, no. 3 (July 1997): 188–223, <https://www.sciencedirect.com/science/article/pii/S1042957397902199> (accessed Feb. 5, 2024).

¹⁸ *Hearing before the United States House of Representatives Committee on Financial Services: Testimony of Vladimir Tenev, Robinhood Markets, Inc.*, 117th Cong. (2021) (testimony of Vladimir Tenev), <https://www.congress.gov/117/meeting/house/111207/witnesses/HHRG-117-BA00-Wstate-TenevV-20210218.pdf> (accessed Feb. 5, 2024).

¹⁹ “Interactive Brokers CEO Welcomes Robinhood to Big Leagues,” *Fox Business*, July 29, 2021, <https://www.foxbusiness.com/markets/interactive-broker-ceo-welcomes-robinhood-to-big-leagues> (accessed Feb. 5, 2024).

some years, PFOF constituted upward of 75% of Robinhood's total revenue.²⁰ Other sources of revenue included subscription services (Robinhood Gold) and interest earned from margin lending.

Table 2. Percentage of total revenue from market makers.

	Year Ending December 31		
	2020	2021	2022
Market Maker			
Citadel Securities	34%	22%	16%
Entities Affiliated with Susquehanna International Group, LLP (1)	18%	12%	8%
Entities Affiliated with Wolverine Holdings, LP (2)	10%	10%	8%
Tai Mo Shan Limited (3)	N/A	15%	3%
All Others Individually Less than 10%	13%	18%	24%
Total as a Percentage of Total Revenue	75%	77%	59%

(1) Consists of Global Execution Brokers, LP and G1X Execution Services, LLC

(2) Consists of Wolverine Execution Services, LLC and Wolverine Securities LLC

(3) Member of Jump Trading Group

Source: Robinhood Markets, Inc., SEC Form 10-K, 2022, <https://www.sec.gov/Archives/edgar/data/1783879/000178387923000045/hood-20221231.htm> (accessed Feb. 6, 2024).

App design

Robinhood also differentiated itself from competitors through app design. The firm's emphasis on design was intended to attract first-time retail investors. Unlike competitors' data-heavy layouts, Robinhood maintained a great deal of white space in its app (**Exhibit 4**), which helped increase reading comprehension.²¹ The app was so inviting, in fact, that Robinhood became the first financial services app to win the Apple Design Award. "There are a bunch of brokerages with ugly stock apps and we're not one of them," noted Joe Binney, Robinhood's vice president of product engineering, in a 2015 interview.²²

The app also used insights from behavioral psychology to gamify investing. "The gamified elements begin as soon as customers start interacting with the app," stated CB Insights.²³ Examples of such design elements included awarding free shares of random stocks for completing certain actions (e.g., referring new customers), sending push notifications to emphasize the volatility of "biggest mover" stocks, and displaying celebratory confetti after placing a trade. Critics contended that these features fed addictive behaviors, similar to video games or casinos.²⁴ "It's a gambling parlor masquerading as a respectable business," quipped Charlie Munger of Berkshire Hathaway Inc.²⁵

²⁰ Jonathan S. Sack and Bronwyn C. Rountree, "Payment for Order Flow and Market Implications," Bloomberg Law, October 2021, <https://www.bloomberglaw.com/external/document/X1RP679S000000/capital-markets-professional-perspective-payment-for-order-flow-> (accessed Feb. 5, 2024).

²¹ "How Top Personal Finance Companies Built the Best PFM Apps," CB Insights, February 10, 2021, <https://www.cbinsights.com/research/personal-finance-apps-strategies/> (accessed Feb. 5, 2024).

²² Josh Constine, "Stock Trading Was Expensive and Ugly. Robinhood's App Makes It Free and Pretty," TechCrunch, October 26, 2015, <https://techcrunch.com/2015/10/26/free-stock-trading-app/> (accessed Feb. 5, 2024).

²³ "How Robinhood Makes Money," CB Insights, July 8, 2021, <https://www.cbinsights.com/research/report/how-robinhood-makes-money/> (accessed Feb. 5, 2024).

²⁴ Michael J. de la Merced and Erin Griffith, "Robinhood Is Fined \$70 Million over Misleading Customers and System Outages," *New York Times*, June 30, 2021, <https://www.nytimes.com/2021/06/30/technology/robinhood-fined-misleading-customers.html> (accessed Feb. 5, 2024).

²⁵ "CNBC Transcript: Billionaire Investors Warren Buffett and Charlie Munger Sit Down with CNBC's Becky Quick for CNBC's 'Buffett & Munger: A Wealth of Wisdom,'" CNBC, June 29, 2021, <https://www.cnbc.com/2021/06/29/cnbc-transcript-billionaire-investors-warren-buffett-charlie-munger-sit-down-with-cnbc-becky-quick-for-cnbc-buffett-munger-a-wealth-of-wisdom-.html> (accessed Feb. 5, 2024).

Gamification arguably encouraged more frequent trading. According to analysts at JPMorgan Chase & Co. (JPMorgan), the average Robinhood user traded 150 times in 2020 compared to 37 times for Schwab users.²⁶ While more frequent trading benefited the company given its revenue model, it did not necessarily benefit users. Indeed, one study found that retail investors who engaged in active trading paid “a tremendous performance penalty.”²⁷ Studies had also found that Robinhood users’ trading was attention induced and forecasted negative future returns.²⁸

Financing and growth

Tenev and Bhatt started pitching Robinhood to potential investors in 2012. The founders struggled to secure funding, facing rejection from approximately 75 venture capital firms.²⁹ Some firms were concerned that Robinhood had yet to launch its platform. Others noted fundamental concerns regarding building a business model around PFOF. “It made me feel bad. Emotionally bad. Because I think it is misleading to people,” stated Billy Gurley, a partner at Benchmark Capital. “My issue with Robinhood is, I think their mission and what they say they stand for is not actually true.”³⁰

In December of 2013, the founders raised \$3 million in seed financing from venture capital firms, including Index Ventures and Andreessen Horowitz.³¹ While the app was still in development, they launched a waitlist to access the beta version. The waitlist quickly went viral, and it gained 50,000 signups in the first week.³² When the app eventually launched at the end of 2014, the waitlist stood at 500,000.³³

In the following years, Robinhood’s product offerings and number of users steadily increased (**Exhibit 5**). While initially users could only trade stocks, the company integrated options and cryptocurrency trading in 2018 and 2019, respectively. Other products launched included fractional shares, cash management services, and Robinhood Gold, a premium subscription service that, among other things, gave users access to professional research and margin trading. Like many other fintech startups, Robinhood funded its growth via venture capital investments, raising over \$500 million within its first five years.³⁴ By 2019, investors valued the company at \$7.6 billion.³⁵

Robinhood benefited from a surge in retail trading during the COVID-19 pandemic. Between 2019 and 2020, the number of users more than doubled, reaching over 12 million. In September of 2020, Robinhood closed its Series G funding round, raising over \$600 million from a group of venture capitalists including Sequoia Capital (Sequoia). “Robinhood has made the financial markets accessible to the masses and, in turn, revolutionized the decades-old brokerage industry,” remarked Andrew Reed, a partner at Sequoia.³⁶ In

²⁶ “Robinhood: Initiating Coverage with an Underweight Rating,” JPMorgan Chase & Co., August 23, 2021.

²⁷ Brad M. Barber and Terrance Odean, “Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors,” *Journal of Finance* 55, no. 2 (April 2000): 773–806, <https://onlinelibrary.wiley.com/doi/abs/10.1111/0022-1082.00226> (accessed Feb. 5, 2024).

²⁸ Brad M. Barber, Xing Huang, Terrance Odean, and Christopher Schwarz, “Attention-Induced Trading and Returns: Evidence from Robinhood Users,” *Journal of Finance* 77, no. 6 (December 2022): 3,141–90, <https://onlinelibrary.wiley.com/doi/full/10.1111/jofi.13183> (accessed Feb. 5, 2024).

²⁹ Sheelah Kolhatkar, “Robinhood’s Big Gamble,” *New Yorker*, May 10, 2021, <https://www.newyorker.com/magazine/2021/05/17/robinhoods-big-gamble> (accessed Feb. 5, 2024).

³⁰ <https://www.newyorker.com/magazine/2021/05/17/robinhoods-big-gamble>.

³¹ Josh Constine, “Robinhood App Will Offer Zero-Commission Stock Trades Thanks to \$3M Seed from Index and A16Z,” TechCrunch, December 18, 2013, <https://techcrunch.com/2013/12/18/zero-commission-stock-trading-robinhood/> (accessed Feb. 5, 2024).

³² Anna Mazarakis and Alyson Shontell, “How Two Founders Got Nearly 1 Million Users for Their App before It Even Existed,” *Business Insider*, July 11, 2017, <https://www.businessinsider.com/free-viral-marketing-how-robinhood-got-1-million-users-before-launch-2017-7> (accessed Feb. 5, 2024).

³³ Josh Constine, “Robinhood Launches Zero-Fee Stock Trading App,” TechCrunch, December 11, 2014, <https://techcrunch.com/2014/12/11/robinhood-free-stock-trading/> (accessed Feb. 5, 2024).

³⁴ Josh Constine, “Stock Trade App Robinhood Raising at \$5B+, Up 4X in a Year,” TechCrunch, March 15, 2018, <https://techcrunch.com/2018/03/15/robinhood-quintacorn/> (accessed Feb. 5, 2024).

³⁵ Kate Clark, “Free Stock Trading App Robinhood Raises \$323M at \$7.6B Valuation,” TechCrunch, July 22, 2019, <https://techcrunch.com/2019/07/22/robinhood-raises-323m/> (accessed Feb. 5, 2024).

³⁶ “Robinhood Raises \$280 Million in Series F Funding Led by Sequoia,” Robinhood Newsroom, May 4, 2020, <https://newsroom.aboutrobinhood.com/robinhood-raises-280-million-in-series-f-funding-led-by-sequoia/> (accessed Feb. 5, 2024).

December, the company hired Goldman Sachs Group, Inc., to lead its IPO, which was planned for the first quarter of 2021.³⁷ The future for Robinhood looked as bright as ever.

Then GameStop happened.

GameStop

Khan was in business school when the meme-stock craze in early 2021 took hold. At the time, she scratched her head at the market behavior, but as she researched, Khan discovered what a critical role meme stocks had in shaping the future of Robinhood. During this episode, multiple stocks that were heavily discussed on social media (e.g., Reddit's r/WallStreetBets) saw their prices temporarily increase to astronomical levels that could not plausibly be explained by their underlying fundamentals. These so-called meme stocks primarily consisted of "struggling companies that somehow caught the popular imagination owing to nostalgia or the desire to root for the underdog."³⁸ Examples included the stocks of AMC Entertainment Holdings Inc., Bed Bath & Beyond Inc., and BlackBerry Ltd. The posterchild of the fad, though, was GameStop Corp. (GameStop).

The popularity of GameStop could largely be attributed to a single person: Keith Gill, a Massachusetts financial adviser with the username "Roaring Kitty" on YouTube.³⁹ In June of 2019, Gill started buying shares of the company at around \$5. At the time, many viewed GameStop, a brick-and-mortar video game retailer, as a company in decline. Indeed, hedge funds and other investors had taken significant short positions in the stock, betting that the stock price would drop.⁴⁰ Retail trading surged during the COVID-19 lockdowns, and Gill's videos garnered increased attention. Some observed that conditions were primed for a short squeeze, where a significant rise in a stock's price would compel short sellers to cover their positions by buying shares, thereby causing the price to increase further.

The short squeeze materialized in January of 2021. At the start of the month, GameStop's share price stood at about \$17. On January 13, shares increased to \$38 following the appointment of Ryan Cohen, the former CEO of Chewy, Inc., to GameStop's board of directors. Over the next two weeks, shares continued to soar, peaking at close to \$380 on January 27. The following day, Gill posted a screenshot on Reddit's r/WallStreetBets showing that his initial \$53,000 investment had ballooned to over \$30 million.⁴¹

Robinhood was the preferred venue for many GameStop traders. On January 26, the brokerage saw a "historical spike" in applications to open accounts after Elon Musk tweeted "Gamestonk!!" Later that week, upward of 700,000 accounts were pending approval.⁴² The surge in trading activity presented challenges for Robinhood. "There are internal things that are starting to buckle under pressure," one engineering manager wrote to a colleague at the height of the frenzy.⁴³

³⁷ "Trading App Robinhood Hires Goldman Sachs to Lead Its IPO, Sources Say," CNBC, December 9, 2020, <https://www.cnbc.com/2020/12/09/trading-app-robinhood-hires-goldman-sachs-to-lead-its-ipo-sources-say-.html> (accessed Feb. 5, 2024).

³⁸ Owen A. Lamont and Richard H. Thaler, "'Dumb Money' Exposes the Baffling Allure of Bad Investment Advice," *New York Times*, October 1, 2023, <https://www.nytimes.com/2023/10/01/opinion/dumb-money-movie-gamestop-stock.html> (accessed Feb. 5, 2024).

³⁹ Julia-Ambra Verlaire and Gunjan Banerji, "Keith Gill Drove the GameStop Reddit Mania," January 29, 2021, <https://www.wsj.com/articles/keith-gill-drove-the-gamestop-reddit-mania-he-talked-to-the-journal-11611931696> (accessed Feb. 5, 2024).

⁴⁰ <https://www.wsj.com/articles/keith-gill-drove-the-gamestop-reddit-mania-he-talked-to-the-journal-11611931696>.

⁴¹ Chris Prentice and Pete Schroeder, "Famed GameStop Bull 'Roaring Kitty' Is a Massachusetts Financial Advisor," Reuters, January 29, 2021, <https://www.reuters.com/article/us-retail-trading-roaringkitty/famed-gamestop-bull-roaring-kitty-is-a-massachusetts-financial-advisor-idUSKBN29Y0AF> (accessed Feb. 5, 2024).

⁴² Charles Forelle, "Employee Scrambled to Keep Robinhood Afloat in January 2021 Meme-Stock Frenzy, House Report Finds," *Wall Street Journal*, June 25, 2022, <https://www.wsj.com/articles/employees-scrambled-to-keep-robinhood-afloat-in-january-2021-meme-stock-frenzy-house-report-finds-11656185206> (accessed Feb. 5, 2024).

⁴³ <https://www.wsj.com/articles/employees-scrambled-to-keep-robinhood-afloat-in-january-2021-meme-stock-frenzy-house-report-finds-11656185206>.

The most pressing issue was collateral. As was standard in the brokerage industry, Robinhood had to post margin (collateral) with a clearinghouse—the National Securities Clearing Corporation (NSCC)—in order to settle trades.⁴⁴ Margin requirements were based on a formula that accounted for the volatility of traded stocks.⁴⁵ The significant activity in GameStop and other meme stocks in late January led to a surge in margin requirements. The situation came to a head the morning of January 28. An article in the *Wall Street Journal* recounted the event:

At 5:11 a.m. on Jan. 28, an automated email from the NSCC landed in a Robinhood inbox: Robinhood was short on its margin by \$3,006,178,364.89. It had until 10 a.m. to fix it. Employees scrambled. Shortly after 6 a.m., a Robinhood operations manager reached a former boss and asked, “Hypothetically what happens if a firm can’t meet their morning NSCC margin settlement?” He provided some advice and an instruction: “wake up your senior leaders.”⁴⁶

Not having sufficient cash to meet what amounted to a tenfold increase in margin requirements, the company scrambled to come up with a solution. First, it raised capital, both drawing down an over \$500 million line of credit from multiple banks and raising over a billion dollars from existing investors.⁴⁷ Second, Robinhood, along with other brokers (e.g., Interactive Brokers), restricted trading in GameStop and other meme stocks. Specifically, on January 28, Robinhood imposed position-closing-only (PCO) restrictions that allowed investors to sell but not buy shares in several companies, including GameStop. The Financial Industry Regulatory Authority (FINRA), the primary regulator for broker-dealers, viewed PCO restrictions “as a risk mitigation tactic to be extraordinary.”⁴⁸

This decision to restrict trading was met with immediate pushback. Rumors circulated on Reddit that Ken Griffin, founder of Citadel Securities, had pressured Robinhood to restrict trading to prevent further losses to short sellers of GameStop.⁴⁹ The rumors were fueled by news reports that Citadel LLC, a hedge fund also founded by Griffin, had provided Melvin Capital, one of the largest GameStop short sellers, with an emergency cash infusion just a few days earlier.⁵⁰ Legislators amplified the claim. “[US House Committee on Financial Services] need to have a hearing on Robinhood’s market manipulation. They’re blocking the ability to trade to protect Wall St. hedge funds, stealing millions of dollars from their users to protect people who’ve used the stock market as a casino for decades,” tweeted congresswoman Rashida Tlaib.⁵¹

⁴⁴ While investors could instantly buy stocks via Robinhood or other brokerages, formal ownership was not immediately transferred. Rather, equity trades in the United States were settled two business days after the execution of a trade (known as T+2). To mitigate counterparty risk (e.g., the buyer not having enough cash), buyers and sellers used clearinghouses that essentially guaranteed the trades.

⁴⁵ In a press release, Robinhood explained the process via which margin requirements were determined: “Clearinghouses look at a firm’s customer holdings as a portfolio. They use a volatility multiplier, looking at specific stocks, to quantify their risk. The clearinghouse may assign significant additional charges based on how much of one stock a firm’s customers hold. If a firm’s customers have more buy than sell orders, and the securities they’re buying are more volatile, the deposit requirement will be higher. Clearinghouses can also require additional deposits if certain thresholds are met.” “What Happened This Week,” Robinhood, Newsroom, January 29, 2021, <https://newsroom.aboutrobinhood.com/what-happened-this-week/> (accessed Feb. 5, 2024).

⁴⁶ <https://www.wsj.com/articles/employees-scrambled-to-keep-robinhood-afloat-in-january-2021-meme-stock-frenzy-house-report-finds-11656185206>.

⁴⁷ <https://www.wsj.com/articles/employees-scrambled-to-keep-robinhood-afloat-in-january-2021-meme-stock-frenzy-house-report-finds-11656185206>.

⁴⁸ “Game Stopped: How the Meme Stock Market Event Exposed Troubling Business Practices, Inadequate Risk Management, and the Need for Legislative and Regulatory Reform,” US Committee on Financial Services, June 2022, https://democrats-financialservices.house.gov/uploadedfiles/6.22_hfsc_gs_report_hmsmeetbp.irm.nlrf.pdf (accessed Feb. 5, 2024).

⁴⁹ Douglas MacMillan and Yeganeh Torbati, “Robinhood and Citadel’s Relationship Comes into Focus as Washington Vows to Examine Stock-Market Moves,” *Washington Post*, January 29, 2021, <https://www.washingtonpost.com/business/2021/01/29/robinhood-citadel-gamestop-reddit/> (accessed Feb. 5, 2024).

⁵⁰ Juliet Chung, “Citadel, Point72 to Invest \$2.75 Billion into Melvin Capital Management,” *Wall Street Journal*, January 25, 2021, <https://www.wsj.com/articles/citadel-point72-to-invest-2-75-billion-into-melvin-capital-management-11611604340> (accessed Feb. 5, 2024).

⁵¹ Rashida Tlaib (@RashidaTlaib), “This is beyond absurd,” Twitter, January 28, 2021, 10:03 a.m., <https://twitter.com/RashidaTlaib/status/1354807292667981828> (accessed Feb. 5, 2024).

Robinhood denied any nefarious intent behind the PCO restrictions. In a blog post on January 29, the company explained its rationale:

It was not because we wanted to stop people from buying these stocks. We did this because the required amount we had to deposit with the clearinghouse was so large—with individual volatile securities accounting for hundreds of millions of dollars in deposit requirements—that we had to take steps to limit buying in those volatile securities to ensure we could comfortably meet our [margin] requirements.⁵²

Regulatory Scrutiny

On February 18, 2021, just three weeks after the meme-stock turmoil, the House of Representatives' Financial Services Committee held the first of three hearings on the matter. The hearing, held virtually, involved “tension, irate questions, people talking over one another, and a cat photo [in Gill’s background].”⁵³ Tenev denied the rumors that third parties had influenced Robinhood’s decision to institute PCO restrictions. “We don’t answer to hedge funds,” he testified. Several committee members raised concerns related to PFOF. One commentator later recounted, “Tenev was also not completely able to assuage the Committee members’ concerns relating to best execution: the duty of brokers to obtain the best possible execution on orders for its [sic] clients.”⁵⁴

The congressional hearing wasn’t the company’s first brush with regulatory ire. In December 2020, for example, the company agreed to pay a fine of \$65 million after the SEC accused it of “failing to satisfy its duty to seek the best reasonably available terms to execute customer orders.”⁵⁵ That same month, Massachusetts regulators filed a complaint against the company, citing “aggressive tactics to attract inexperienced investors, its use of gamification strategies to manipulate customers, and its failure to prevent frequent outages and disruptions on its trading platform.”⁵⁶

Scrutiny by regulators seemed to have increased after GameStop. In July of 2021, FINRA fined Robinhood \$70 million. The fine, the largest in FINRA’s history, was in response to false and misleading information provided to customers as well as system outages at Robinhood in 2020. PFOF was also subject to increased scrutiny. SEC chair Gary Gensler led the charge. In June 2021, Gensler stated that Robinhood “explicitly offered to accept less price improvement for its customers in exchange for receiving higher payment for order flow for itself. As a result, many Robinhood customers shouldered the costs of inferior executions.”⁵⁷ Gensler noted that it wasn’t clear whether the costs from inferior execution were lower than the savings from eliminating commissions.

⁵² <https://newsroom.aboutrobinhood.com/what-happened-this-week/>.

⁵³ Pete Schroeder, Svea Herbst-Bayliss, and John McCrank, “Long, Tense with Cat Photo for Relief; How the GameStop Hearing Unfolded,” Reuters, February 18, 2021, <https://www.reuters.com/article/us-retail-trading-usa-congress/long-tense-with-cat-photo-for-relief-how-the-gamestop-hearing-unfolded-idUSKBN2A11C4> (accessed Feb. 5, 2024).

⁵⁴ Edward Segal, “Robinhood’s CEO Receives Mixed Review for Congressional Testimony About App’s Role in GameStop Saga,” *Forbes*, February 19, 2021, <https://www.forbes.com/sites/edwardsegal/2021/02/19/robinhoods-ceo-receives-mixed-reviews-for-congressional-testimony-about-apps-role-in-gamestop-saga/?sh=18e1e66b3e63> (accessed Feb. 5, 2024).

⁵⁵ “SEC Charges Robinhood Financial with Misleading Customers about Revenue Sources and Failing to Satisfy Duty of Best Execution,” United States Securities and Exchange Commission, December 17, 2020, <https://www.sec.gov/news/press-release/2020-321> (accessed Feb. 5, 2024).

⁵⁶ Pippa Stevens, “Massachusetts Regulators File Complaint against Robinhood, Alleging Manipulation of Customers,” CNBC, December 16, 2020, <https://www.cnbc.com/2020/12/16/robinhood-reportedly-facing-complaint-by-massachusetts-regulators-over-targeting-young-users.html> (accessed Feb. 5, 2024).

⁵⁷ “Testimony before the House Committee on Financial Services,” United States Securities and Exchange Commission, May 6, 2021, <https://www.sec.gov/news/testimony/gensler-testimony-20210505> (accessed Feb. 5, 2024).

The extent to which the SEC would address PFOF was an open question. In August of 2021, Gensler stated that a ban on PFOF was “on the table.”⁵⁸ Analysts at JPMorgan called the risk of a PFOF ban remote, giving it a probability of between 1% and 3%.⁵⁹ Perhaps more likely was that the SEC would either cap fees or mandate additional disclosures. Another possibility analysts raised was that a competitor such as Schwab “might eliminate PFOF in a predatory manner to pressure Robinhood’s business model.” Robinhood’s plan B in such a scenario might be to become a market maker itself, but it was unclear whether such a move would be approved by the SEC.⁶⁰

IPO and Subsequent Performance

Despite the turmoil, Robinhood’s growth had initially not appeared to stall. **Table 3** compares key operating metrics for the first quarter of 2020 and 2021.⁶¹ The company decided to proceed with its previously planned IPO.

Table 3. Key operating metrics, 2020 versus 2021.

<i>In millions except ARPU</i>	Period Ending March 31	
	2020	2021
Net Cumulative Funded Accounts	7.2	18.0
Monthly Active Users (MAU)	8.6	17.7
Assets Under Custody (AUC)	\$19,220.1	\$80,932.4
Average Revenues Per User (ARPU)	\$82.9	\$137.0

Source: Robinhood SEC Form S-1, 2021, <https://www.sec.gov/Archives/edgar/data/1783879/000162828021013318/robinhoods-1.htm> (accessed Feb. 6, 2024).

Robinhood pledged to make 35% of its IPO shares available to its users. The company planned to use its new service, IPO Access, to allocate shares. IPO Access allowed Robinhood customers to purchase shares at the IPO price. Typically, IPO shares were sold through a syndicate of investment bankers who, in turn, sold them to large investors such as hedge funds or mutual funds. Retail investors wanting to buy shares of a newly public company had to wait until the stock opened for trading, often buying shares at a significant markup to the IPO price. Robinhood’s own IPO was among the first to use the service. In the next year, several other IPOs (e.g., Sweetgreen, Inc., and Allbirds, Inc.) would use it as well.⁶²

Robinhood shares made their Nasdaq debut on July 29, 2021. The stock opened at \$38, implying a market cap for the company of over \$30 billion. Tenev and Bhatt each owned stakes valued at about \$2.4 billion.⁶³ While the stock struggled that first day, closing below \$35, it skyrocketed during the next week, reaching a high of over \$70 on August 4. The good times did not last. By the end of October, shares had fallen back below \$40. In the first quarter of 2022, shares collapsed to around \$10 and remained at this level for the rest of the year (**Exhibit 1**).

⁵⁸ Avi Salzman, “SEC Chairman Says Banning Payment for Order Flow Is ‘On the Table,’” *Barron’s*, August 30, 2021, <https://www.barrons.com/articles/sec-chairman-says-banning-payment-for-order-is-on-the-table-51630350595> (accessed Feb. 5, 2024).

⁵⁹ “Robinhood: Initiating Coverage with an Underweight Rating.”

⁶⁰ “Robinhood: Initiating Coverage with an Underweight Rating.”

⁶¹ Robinhood Markets, Inc. SEC Form S-1, 2021, <https://www.sec.gov/Archives/edgar/data/1783879/000162828021013318/robinhoods-1.htm> (accessed Feb. 5, 2024).

⁶² Annie Massa and Mathieu Benhamou, “Robinhood Gave Its Customers Access to IPOs that All Flopped,” *Bloomberg*, November 8, 2022, <https://www.bloomberg.com/news/articles/2022-11-08/robinhood-gave-its-customers-access-to-ipos-that-all-flopped> (accessed Feb. 5, 2024).

⁶³ Peter Rudegair, Corrie Driebusch, and Caitlin McCabe, “Robinhood’s Stock Price Falls after IPO,” *Wall Street Journal*, July 29, 2021, <https://www.wsj.com/articles/robinhood-stock-opens-flat-in-trading-debut-11627576759> (accessed Feb. 5, 2024).

The falling share price reflected weakness in the underlying business. The surge in users in early 2021 had proven to be transitory. By the middle of 2022, both assets under custody (AUC) and monthly active users (MAU) had dropped over a third from their highs (**Exhibit 6**). Revenues also exhibited a steep drop-off, and the company lost over a billion dollars in 2022 (**Exhibit 7**).

The Future of Robinhood

In 2022, the year Khan started her current position, Robinhood rolled out new services to mitigate the effects of lower transaction revenues. “Explosive growth in 2020 and 2021 had consumed many resources that otherwise would have been invested in longer-term projects,” explained an article in the *Wall Street Journal*. Tenev noted that the goal was to fill “some of the basic product gaps in the investing offering.”⁶⁴ To this end, Robinhood started offering retirement accounts (i.e., traditional and Roth IRAs) in 2022. The company also planned to offer brokerage services in the United Kingdom by the end of the following year.⁶⁵

Still, the company’s future was subject to considerable uncertainty. Some speculated that Robinhood was a takeover target. Indeed, its competitors TD Ameritrade and E-Trade had already been acquired by Schwab and Morgan Stanley, respectively. One natural suitor appeared to be Sam Bankman-Fried, the founder and CEO of the cryptocurrency exchange FTX Trading Ltd. (FTX). In May of 2022, Bankman-Fried acquired a 7.6% stake in Robinhood, saying at the time, “There is a really plausible world in which it has a huge comeback.”⁶⁶ The collapse of FTX that November and fraud allegations against Bankman-Fried dashed any hopes of a potential acquisition, however.

Some thought Robinhood had the potential to become much more than a broker. Perhaps the value of Robinhood customers was not just the fees generated from their portfolios but the opportunity to provide additional products and services to them in the future. The unique demographics of Robinhood users provided an advantage in this regard. “Robinhood has something no one else has: 22.5 million youngsters. That’s huge,” noted Thomas Peterffy, the founder and former CEO of Interactive Brokers.⁶⁷ An article from Dealbook noted that Robinhood envisioned opportunities beyond the investment space: “If Robinhood can retain its young customers through to retirement, or better yet convert them into credit card or banking customers, as Robinhood says it wants to, those accounts could be worth a lot more over time.”⁶⁸

Other fintech companies had expanded their services in a race to create a one-stop shop for financial services (**Exhibit 8**). One commentator referred to this as “the great rebundling”—fintech companies that had previously unbundled financial services were in the process of bundling them back together.⁶⁹ A January 2022 article from Bloomberg noted that peer-to-peer payment platforms had gone a long way toward achieving this goal.⁷⁰ Could Robinhood catch up? Some analysts thought so, noting, “Every company in fintech is chasing the Alipay ‘super-app’ concept...we think HOOD [Robinhood] stands about as good a chance to get there.”⁷¹

⁶⁴ Peter Rudegeair, “Robinhood Wants Your Retirement Money,” *Wall Street Journal*, December 6, 2022, <https://www.wsj.com/articles/robinhood-wants-your-retirement-money-11670300662> (accessed Feb. 5, 2024).

⁶⁵ “Robinhood Markets, Inc. Earnings Presentation Fourth Quarter 2022,” Robinhood, February 8, 2023, https://s28.q4cdn.com/948876185/files/doc_financials/2022/q4/Q4-2022-Earnings-Presentation-vDISTRO-FINAL.pdf (accessed Feb. 5, 2024).

⁶⁶ Caitlin McCabe, Gunjan Banerji, and Alexander Osipovich, “The Unraveling of Robinhood’s Fairy Tale,” *Wall Street Journal*, June 18, 2022, <https://www.wsj.com/articles/the-unraveling-of-robinhoods-fairy-tale-11655524803> (accessed Feb. 5, 2024).

⁶⁷ Andrew Ross Sorkin, Jason Karaian, Sarah Kessler, Stephen Gandel, Lauren Hirsch, and Ephrat Livni, “What Is Robinhood Worth?” *New York Times*, July 29, 2021, <https://www.nytimes.com/2021/07/29/business/dealbook/robinhood-ipo-valuation-worth.html> (accessed Feb. 13, 2024).

⁶⁸ <https://www.nytimes.com/2021/07/29/business/dealbook/robinhood-ipo-valuation-worth.html>.

⁶⁹ Medha Agarwal, “What’s Old Is New: The Great Rebundling,” Medium, September 6, 2022, <https://medhaa.medium.com/whats-old-is-new-the-great-rebundling-a32ec6f7a7e8> (accessed Feb. 7, 2024).

⁷⁰ Julie Chariell and Mark Glassman, “Startups from Chime to Robinhood Race to Build a Money Super App,” Bloomberg, January 14, 2022, <https://www.bloomberg.com/news/articles/2022-01-14/fintech-startups-from-chime-to-robinhood-race-to-build-a-money-super-app> (accessed Feb. 5, 2024).

⁷¹ “Looks Good Under the HOOD; Initiate at EW,” Barclays Capital Inc., August 23, 2021.

Analysts at JPMorgan expressed skepticism regarding whether such a goal could be achieved, noting that the company faced formidable competition.⁷²

The Decision

The night before Khan planned to make a recommendation to Moore, she compiled the information she had read and turned to a fresh sheet in her notebook to jot down some thoughts. Many investors had written off Robinhood in response to its recent struggles. But Khan prided herself on seeing opportunities where others did not. Could a turnaround be in the cards for the company? Or was the goal of a new Wall Street ultimately unobtainable?

⁷² “Robinhood: Initiating Coverage with an Underweight Rating.”

Exhibit 1

Robinhood: Welcome to the New Wall Street?

Robinhood Markets, Inc., Share Price (in US dollars)

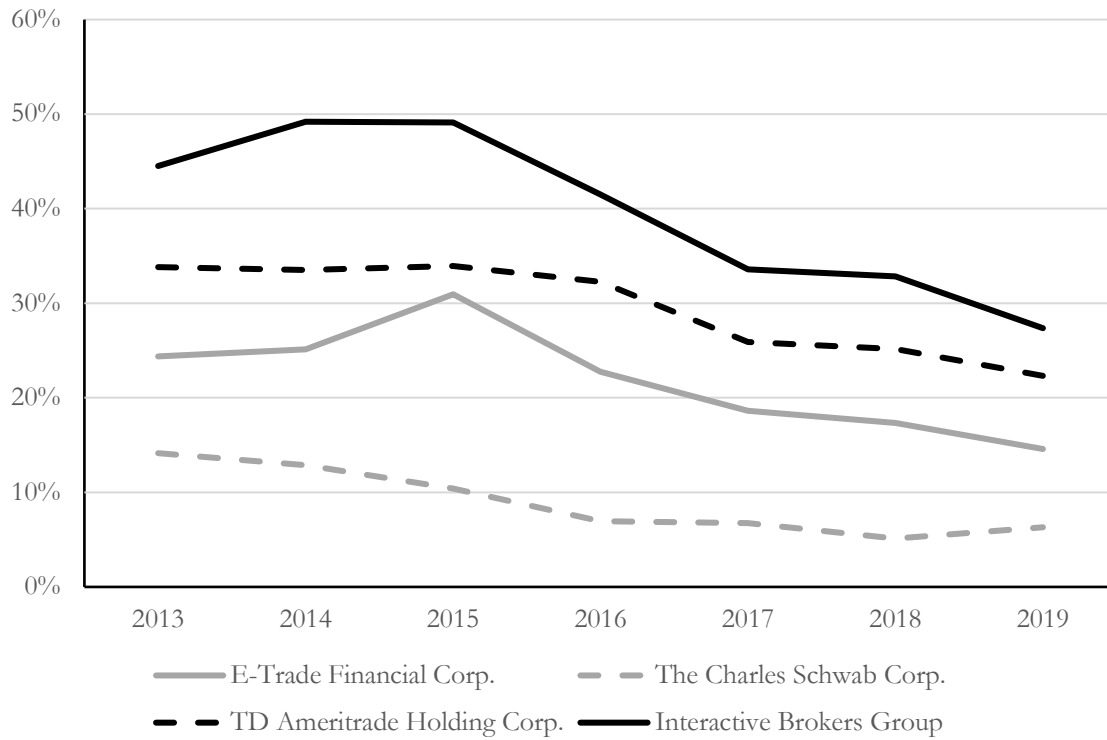


Source: Created by authors using Nasdaq share price listings for 2021–2023.

Exhibit 2

Robinhood: Welcome to the New Wall Street?

Discount Broker Commissions as a Percentage of Revenues (2013–2019)

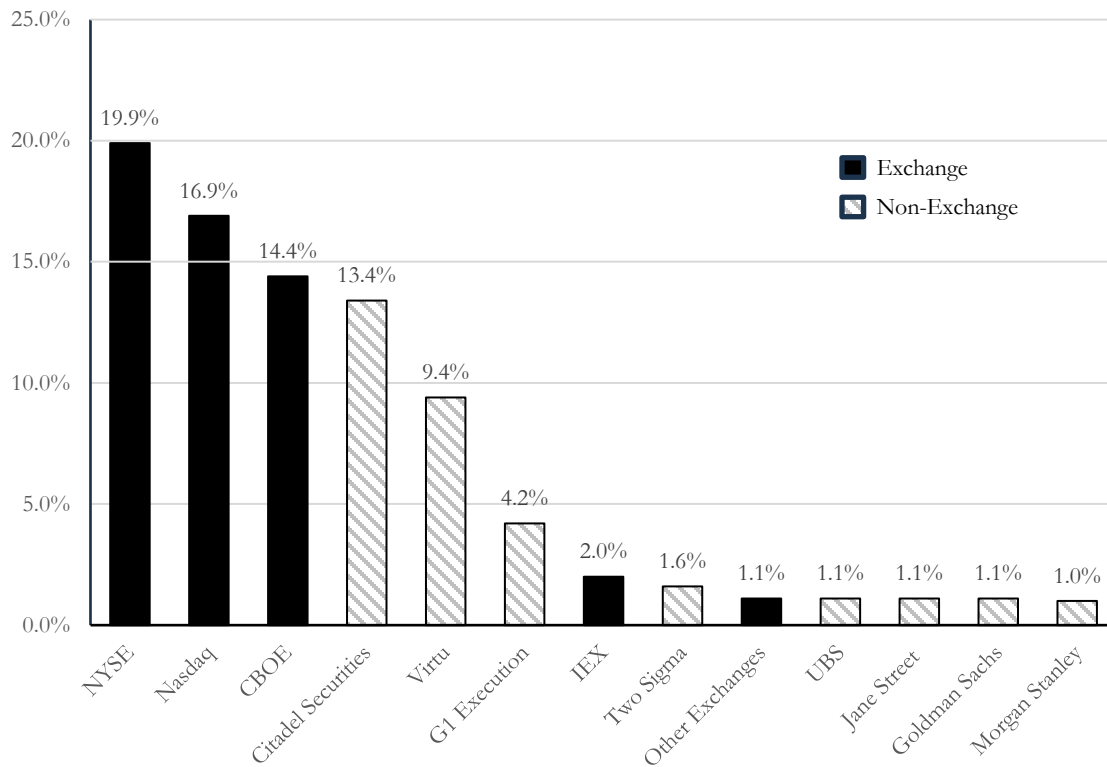


Data source: SEC Forms 10-K for all companies and years presented in this exhibit.

Exhibit 3

Robinhood: Welcome to the New Wall Street?

Trading Venues (December 2020)

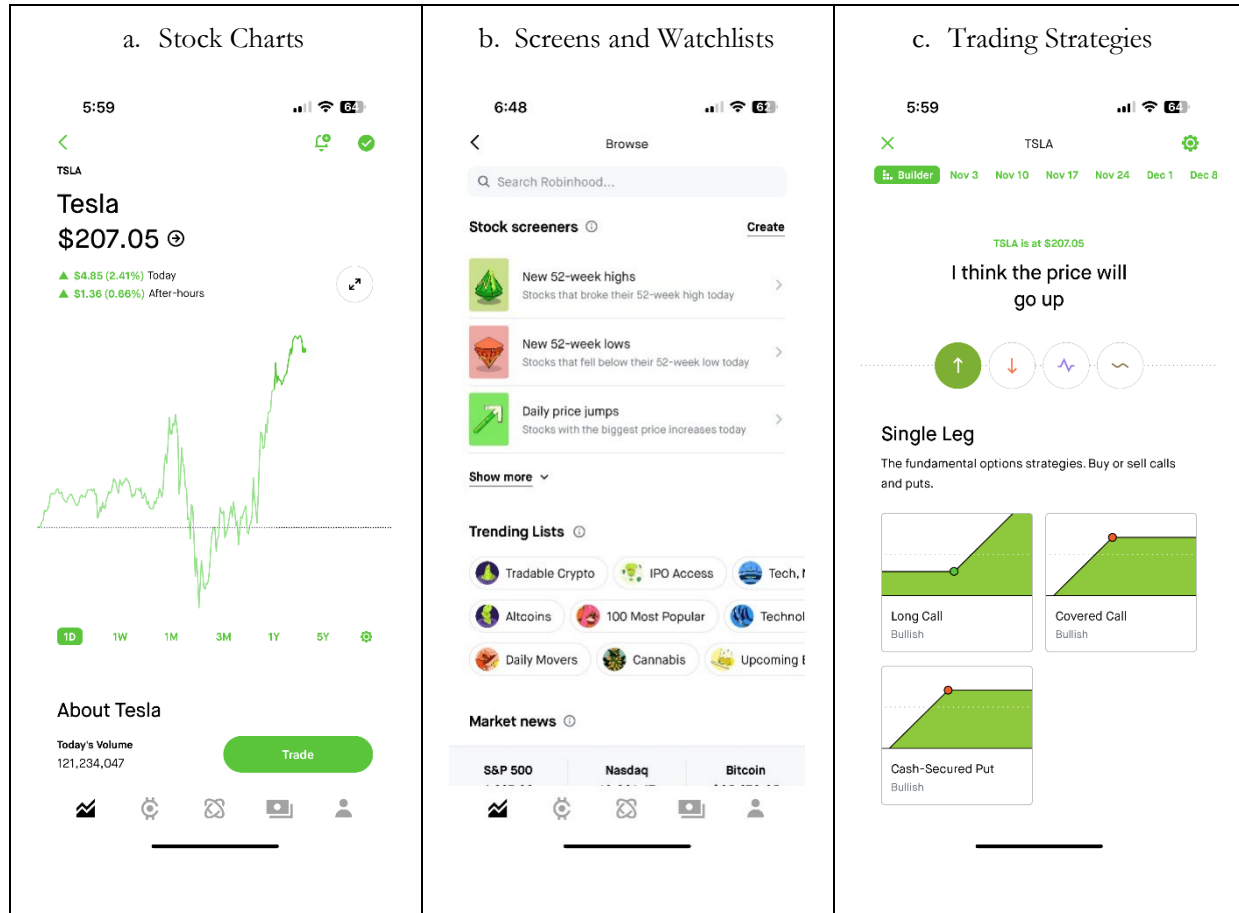


Data source: John Detrixhe, "Citadel Securities Gets almost as Much Trading Volume as Nasdaq," Quartz, February 5, 2021, <https://qz.com/1969196/citadel-securities-gets-almost-as-much-trading-volume-as-nasdaq> (accessed Feb. 15, 2024).

Exhibit 4

Robinhood: Welcome to the New Wall Street?

Interface

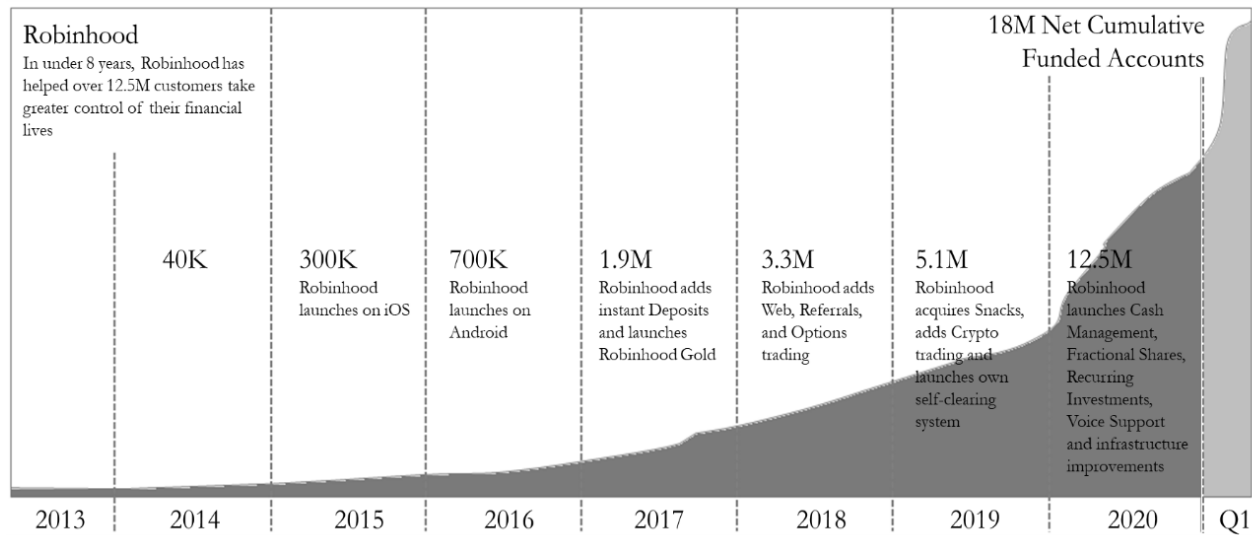


Source: From author's own Robinhood account.

Exhibit 5

Robinhood: Welcome to the New Wall Street?

Robinhood Markets, Inc., User Growth and Product Launches



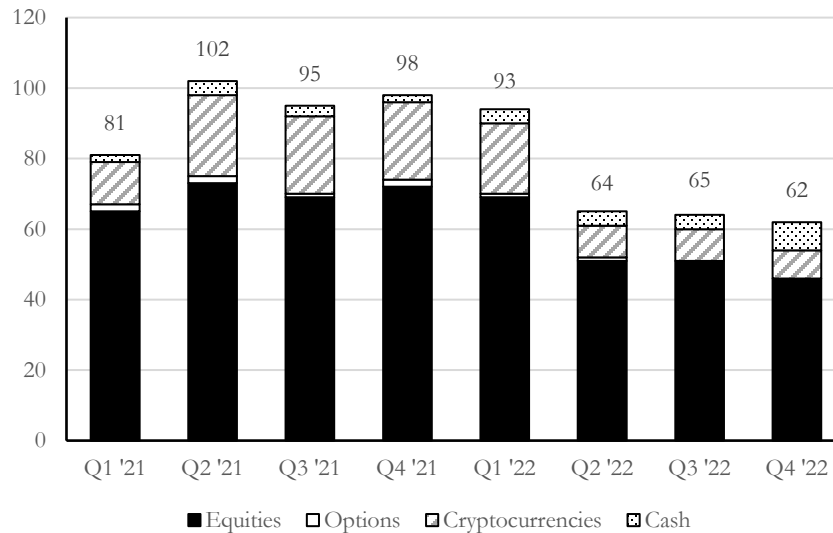
Source: Robinhood Markets, Inc. SEC Form S-1, 2021, <https://www.sec.gov/Archives/edgar/data/1783879/000162828021013318/robinhoods-1.htm> (accessed Feb. 5, 2024).

Exhibit 6

Robinhood: Welcome to the New Wall Street?

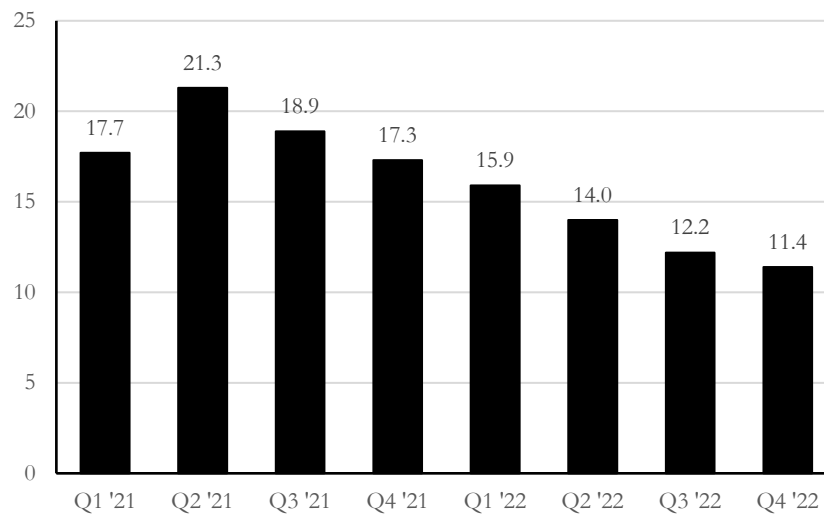
Assets under Custody and Monthly Active Users, 2020–2022

Panel A: Assets under Custody (in billions of US dollars)



Data source: “Robinhood Markets, Inc. Earnings Presentation Second Quarter 2023,” Robinhood, August 2, 2023, https://s28.q4cdn.com/948876185/files/doc_financials/2023/q2/Q2-2023-Earnings-Presentation-vE.pdf; “Robinhood Markets, Inc. Earnings Presentation First Quarter 2023,” Robinhood, May 10, 2023, https://s28.q4cdn.com/948876185/files/doc_financials/2023/q1/Q1-2023-Earnings-Presentation.pdf (both accessed Feb. 5, 2024).

Panel B: Monthly Active Users (in millions)



Data source: “Robinhood Markets, Inc. Earnings Presentation Second Quarter 2023,” Robinhood, August 2, 2023, https://s28.q4cdn.com/948876185/files/doc_financials/2023/q2/Q2-2023-Earnings-Presentation-vE.pdf; “Robinhood Markets, Inc. Earnings Presentation First Quarter 2023,” Robinhood, May 10, 2023, https://s28.q4cdn.com/948876185/files/doc_financials/2023/q1/Q1-2023-Earnings-Presentation.pdf (both accessed Feb. 5, 2024).

Exhibit 7

Robinhood: Welcome to the New Wall Street?

Robinhood Markets, Inc., Income Statement, 2020–2022

	Year Ending December 31,		
	2020	2021	2022
Revenues			
Transaction-Based Revenues	720.0	1,402.0	814.0
Net Interest Revenue	177.0	256.0	424.0
Other Revenues	61.0	157.0	120.0
Expenses			
Operations	(135.0)	(368.0)	(285.0)
Brokerage Transactions	(114.0)	(158.0)	(179.0)
Marketing	(186.0)	(325.0)	(103.0)
General and Administrative	(295.0)	(1,371.0)	(924.0)
Technology Development	(215.0)	(1,234.0)	(878.0)
Other Expenses (Net)	0.0	(2,044.0)	(16.0)
Earnings before Taxes	13.0	(3,685.0)	(1,027.0)
Provision for Income Tax	(6.0)	(2.0)	(1.0)
Net Income	7.0	(3,687.0)	(1,028.0)

Data source: Capital IQ.

Exhibit 8

Robinhood: Welcome to the New Wall Street?

Financial Services Offered by Fintech Companies (as of January 10, 2022)

● Current offering ○ Planned offering	<u>Digital Wallet Apps</u>				<u>Investing/Banking</u>			
	PayPal	Venmo	Cash App	Shop Pay	SoFi	Chime	Coinbase	Robinhood
Store cash balance and cards	●	●	●	●	●	●	●	●
QR code payments	●	●						
E-commerce payment button	●	●		●				
Peer-to-peer transfers	●	●	●		●	●		
Instant transfers to bank accounts	●	●	●				●	●
Debit card	●	●	●		●	●	●	
Credit card	●	●			●	●		
Checking account	●	●	●		●	●		
Savings account	●				●	●		●
Bill payment	●				●	●		
Direct deposit	●	●	●		●	●	●	
Buy now, pay later	●		○	●				
Consumer loans	●		●		●			
Cryptocurrency trading	●	●	●		●		●	●
Stock trading	○	○	●		●			●
Budgeting/credit building					●	●		
Shopping deals	●	●	●	●				
Loyalty program/rewards	●	●	●		●		●	●
Monthly active users (in millions)	85	33	36	10	1	7	21	16

Data source: Julie Chariell and Mark Glassman, "Startups from Chime to Robinhood Race to Build a Money Super App," Bloomberg, January 14, 2022, <https://www.bloomberg.com/news/articles/2022-01-14/fintech-startups-from-chime-to-robinhood-race-to-build-a-money-super-app> (accessed Feb. 5, 2024).